

Pradeep Pandey

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SUMMARY

Computer Science junior with an Applied Mathematics minor, focused on software engineering, simulation tools, full-stack applications, and database-backed systems. Built GUI, web, cloud, and systems projects using C#, .NET, C++, Python, React, FastAPI, Flask, SQL, and AWS. Seeking 2026 software engineering, automation, test systems, or full-stack development internships.

EDUCATION

University of Wisconsin–Stout

Menomonie, WI

B.S. Computer Science, Minor in Applied Mathematics; GPA: 3.43/4.0

Expected Dec. 2026

- Dean's List: Spring 2023, Spring 2025, Fall 2025
- Relevant Courses: Software Engineering Practicum, Database Systems, Algorithms, Operating Systems, Cryptography, Formal Languages and Automata, Statistics, Computer Architecture, Discrete Mathematics, Linear Algebra, Calculus

TECHNICAL SKILLS

Languages: C#, C++, Python, Java, JavaScript, TypeScript, SQL, PHP, C

Frameworks: .NET 8, WinForms, React, Vite, FastAPI, Flask, Node.js, Tailwind, Redux

Databases: SQLite, MySQL, AWS RDS, schema design, data persistence, query debugging

Tools: Visual Studio, VS Code, Git/GitHub, CMake, SFML, Linux/Ubuntu, Nginx, Gunicorn, Postman

SOFTWARE EXPERIENCE

Open Source Software Developer, IIIF 3D Manifest Editor

2026 – Present

React, TypeScript, Vite, Tailwind, JSON, IIIF Presentation API

github.com/Wilsont0554/ManifestEditor

- Working on a 3D viewing and manifest editing website for IIIF resources, helping users create, edit, preview, and download structured 3D manifests using IIIF 4.0 API.
- Built editor workflows for adding 3D models, images, cameras, lights, annotations, metadata, and transform controls.
- Implemented JSON import/export, live manifest preview, resource editing, and UI improvements for non-technical users.
- Collaborated in Agile Scrum sprints by writing user stories, sprint goals, implementation notes, and confirmation criteria.

PROJECTS

Fiber Optic Alignment and Test Station Simulator | C#, .NET 8, WinForms, SQLite

2026

- github.com/pandeyp1426/Fiber-Optic-Alignment-and-Test-Station-Simulator
- Built a personal desktop simulator project for a fiber optic alignment and test station with a WinForms GUI, simulated sensor readings, and test result tracking.
- Simulated X/Y alignment, optical power, and temperature readings, then validated each run against pass/fail tolerance rules.
- Stored completed test runs, result status, and failure reasons in SQLite so previous test history could be reviewed inside the application.

RankMyStocks | React, Vite, Redux, Auth0, Flask, MySQL, AWS EC2/RDS

2025

- rankmystocks.com | github.com/pandeyp1426
- Developed and deployed a full-stack stock discovery and portfolio ranking app with React, Flask REST APIs, Auth0 login, MySQL persistence, and AWS cloud hosting.
- Designed portfolio, transaction, and snapshot tables, migrated persistence to AWS RDS, and deployed the backend on EC2 using Ubuntu, Nginx, and Gunicorn.

WFC Biome Generator | C++, SFML, CMake, OOP, Procedural Generation

2026

- github.com/pandeyp1426/WFCBiomeGenerator
- Built a C++/SFML desktop app that generates biome maps using Wave Function Collapse-style constraint propagation and DFA-based adjacency rules.
- Structured biome rules, cells, entropy updates, rendering, and regeneration controls using object-oriented design and CMake project setup.

Cryptography CTF Challenges Workspace | React, Vite, FastAPI, Python, SQLite, RSA

2026

- github.com/pandeyp1426/Cryptography_CTF_CHALLENGES
- Built a full-stack CTF solving workspace with challenge input, file upload parsing, streamed solve steps, progressive hints, flag detection, and SQLite solve history.
- Implemented RSA helper logic for shared prime attacks, modular inverse, GCD-based factoring, and classroom-style crypto writeups.

LEADERSHIP & CREDENTIALS

Event Manager and AWS Core Team Member, UW–Stout AWS Club

Dec. 2025 – Present

Cloud computing workshops, student mentoring, event coordination

Menomonie, WI

- Coordinate with club members to teach and support students learning AWS services, Terraform, EC2, S3, RDS, cloud deployment, and related cloud computing tools.
- Help plan technical workshops, promote club events, organize room logistics, and create hands-on learning opportunities for students interested in cloud computing.

Credentials: AWS Cloud Club Core Team Badge, AWS Community, Dec. 2025 | [Google Data Analytics Foundations](https://www.google.com/certificates/cloud-foundations/), Coursera, Mar. 2025